A 442.42 TOPS/W RRAM-based Digital Computing-in-Memory Accelerator for BF16×1-bit Vision Transformer

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***Abstract*—RRAM-based Computing-in-Memory (CIM) architectures are promising for efficient Transformer inference, yet analog approaches suffer from circuit non-idealities and limited precision, leading to accuracy degradation. In this paper, a fully digital CIM accelerator based on RRAM is proposed to achieve a better trade-off between accuracy, energy, and area. Firstly, a hardware-oriented W1A16 quantization strategy is proposed, incorporating the architectural enhancements of BinaryViT to compensate for the capacity loss of 1-bit weights. By maintaining activations in Brain-Floating Point-16 (BF16) format and utilizing hard distillation, the quantized Vision Transformer achieves competitive accuracy with minimized memory footprint. Then, a fully digital RRAM-based CIM accelerator is designed with an area-efficient 4T-XOR and a fine-grained mantissa shifter. Implemented on 28nm technology node, the proposed design achieves a peak normalized energy efficiency of 442.42 TOPS/W.**

***Index Terms*—Computing-in-Memory, RRAM, ViT, Quantization, Energy-efficient.**

I. Introduction

The advancement of large language models (LLM) like Transformer poses significant challenges to traditional von Neumann architectures, primarily due to the memory wall, bandwidth demands, and prohibitive energy costs associated with frequent data movement [1] [2]. CIM architecture reduces the need for data movement by facilitating data processing within memory [3], which is recognized as a promising candidate for efficient computation of LLM [4] [5] [6].

Despite the exceptional image processing performance of Vision Transformers (ViTs) [7], [8], their large size impedes edge deployment. Although quantization compresses models [9], extreme low-bit quantization severely harms accuracy [10], [11].

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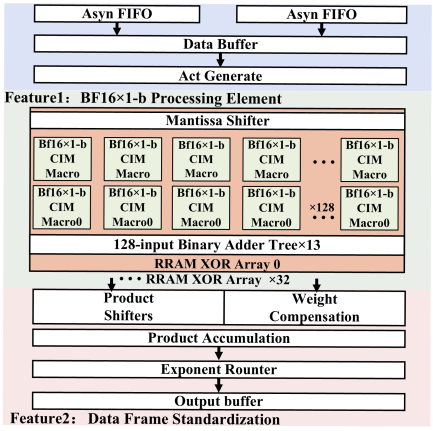


Fig. 1. BF16×1-bit RRAM based CIM accelerator for ViT.

RRAM is a promising candidate for CIM architectures due to its compact cell size, non-volatility, and high energy efficiency [12]–[14]. However, analog RRAM-based CIM schemes often sacrifice precision and flexibility for low power consumption [15]–[17], and typically require serial input of multi-bit data [18], leading to frequent voltage stress that degrades device reliability [19]. While digital CIM accelerators improve accuracy [20]–[22], they often suffer from overheads caused by complex adder trees and cell interconnections.

As illustrated in Fig. 1, the proposed design features a hierarchical structure of PE Groups, PEs, and RRAM arrays. Each PE Group shares activations across 32 PEs, while each PE integrates parallel RRAM arrays with a digital adder tree and a shared readout circuit. Key innovations introduced to address prior limitations include:

1) **Hardware-Oriented W1A16 Quantization:** A hardware-oriented quantization strategy is proposed

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utilizing 1-bit weights and BF16 activations to maximize RRAM storage density while preserving the high dynamic range required by attention mechanisms.

2) **RRAM-based BF16×1-bit CIM Macro:** We propose a novel CIM architecture that supports BF16×1-bit computation, achieving high throughput and reconfigurability while maintaining energy efficiency.
3) **Area-Accuracy Co-design:** Our implementation incorporates area-efficient 4T-XOR logic combined with a batch-wise mantissa shift scheme, effectively balancing silicon area and computational accuracy.

This paper is structured as follows. Section II introduces binarized ViT with knowledge distillation. Section III presents the proposed design of the hardware system. Section IV shows the evaluation results. Section V concludes this paper.

II. Hardware-Oriented Quantization and Architecture

To maximize the energy efficiency of the proposed RRAM-based CIM accelerator while maintaining high inference accuracy, we propose a hardware-algorithm co-design strategy. Unlike prior works that pursue extreme 1-bit quantization for both weights and activations (W1A1), we adopt a **W1A16** (1-bit Weight, BF16 Activation) quantization scheme. This strategy leverages the high storage density of RRAM for binary weights while utilizing the digital CIM macro for high-precision BF16 accumulation, addressing the dynamic range issues often found in fully binarized Vision Transformers (ViTs).

A. Architecture Optimization

Standard ViTs, such as DeiT, suffer significant accuracy degradation under binarization due to limited representational capability. To mitigate this, we adopt and scale down the advanced architectural topology proposed in BinaryViT [23] to the Tiny scale. We incorporate three key operations to enrich information flow without introducing heavy convolutions:

1) **Global Average Pooling:** We replace the standard token pooling with global average pooling to aggregate information from all tokens before the classifier.
2) **Multi-Branch Feed-Forward Network:** We utilize multiple average pooling branches in the FFN to compensate for the information loss caused by weight binarization.
3) **Pyramid Structure:** A hierarchical feature map structure is employed to process features at higher resolutions in early stages, enhancing feature extraction capabilities.

B. W1A16 Quantization with Hard Distillation

Weight Binarization: Weights in all linear layers (Attention and MLP) are binarized to $\{-1, +1\}$ to minimize memory footprint. We utilize a center-mean sign function with a scaling factor to minimize quantization error:

$$B_w = \alpha \cdot \text{sign}(W - \bar{W}), \quad \alpha = \frac{1}{n}\|W - \bar{W}\|_1 \tag{1}$$

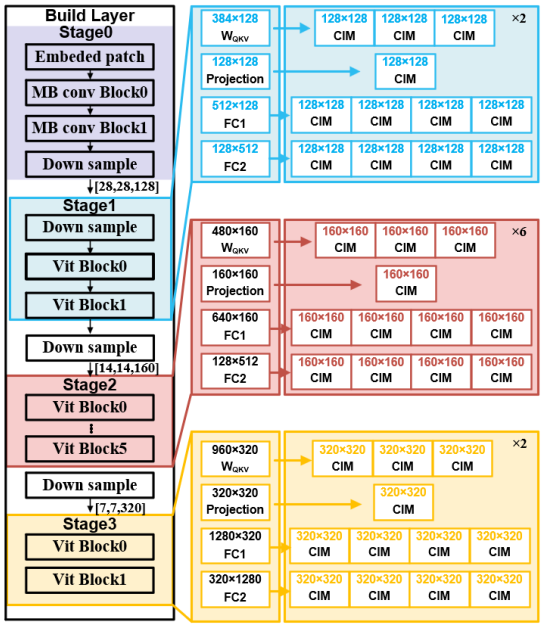


Fig. 2. RRAM-based CIM array architecture for ViT.

where $\bar{W}$ denotes the mean of the weight matrix, and α is the layer-wise scaling factor. This allows 1-bit storage directly in RRAM cells.

BF16 Activations: Activations are maintained in Brain-Floating Point 16 (BF16) format. Unlike 1-bit activations, BF16 preserves the high dynamic range essential for the attention mechanism (Softmax and GELU layers), ensuring the model captures subtle visual features that are typically lost in W1A1 models.

Hard Distillation: To recover accuracy, we employ a hard distillation strategy [24]. Using a full-precision DeiT-Tiny as the teacher, the student model learns to mimic the teacher's hard decisions. The total loss function $\mathcal{L}_{\text{total}}$ is defined as:

$$\mathcal{L}_{\text{total}} = \frac{1}{2}\mathcal{L}_{\text{CE}}(\psi(Z_s), y) + \frac{1}{2}\mathcal{L}_{\text{CE}}(\psi(Z_s), y_t) \tag{2}$$

where Z_s is the student logits, y is the ground truth, $y_t = \text{argmax}(Z_t)$ is the hard label from the teacher, and ψ is the softmax function. This approach forces the binary weights to converge towards an optimal configuration that effectively emulates the full-precision model.

III. The Architecture of Proposed CIM Accelerator

A. RRAM-based BF16×1-bit CIM PE

In the ViT computational flow, some operands are dynamic, making writes to RRAM-CIM a bottleneck due to high latency. To address this, we reserve RRAM-CIM solely for storing the stable, quantized network weights (approximately 4 Mb) shown in Fig. 2, leveraging its strength in high-throughput, energy-efficient read operations. The three Stages involve matrices of different dimensions. Therefore, the CIM arrays are designed

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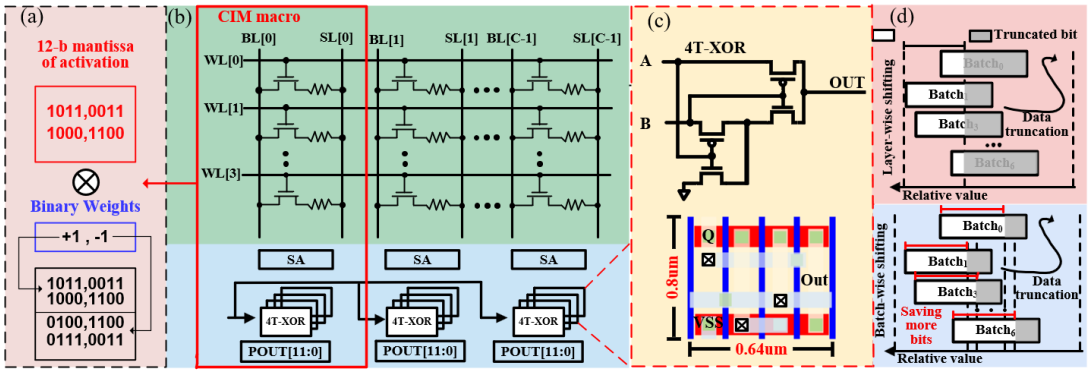


Fig. 3. (a) Functionality of the BF16×1-bit CIM macro (b) The structure of the proposed digital RRAM CIM array. (c) Architecture of the area-optimized 4T-XOR and layout. (d) Comparison between layer-wise shifting and batch-wise shifting.

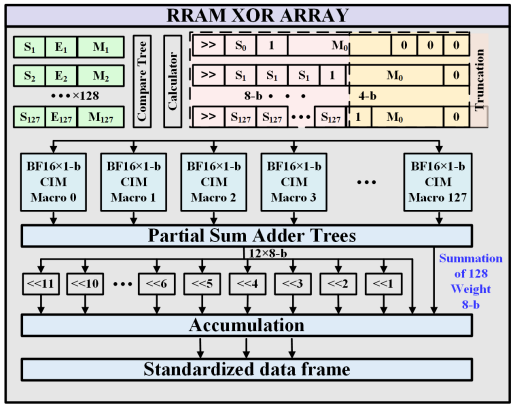


Fig. 4. RRAM-based CIM PE for BF16×1-bit computation.

in three sizes: 128×128, 160×160, and 320×320. Taking the 128×128 case as an example, as illustrated in Fig. 4, the CIM PE integrates 32 parallel CIM arrays. Each of these arrays comprises 128 CIM macros, with each macro containing four 1T1R computing cells arranged in the same column. To support exponent pre-alignment and improve computational accuracy, a fine-grained mantissa shifter is incorporated within the architecture. Following exponent alignment, the PE performs high-precision multiply-accumulate (MAC) operations using 12-bit mantissas.

B. RRAM-based BF16×1-bit CIM Array

As shown in Fig. 3(b), the proposed RRAM-based CIM macro employs a 1T1R structure, with each column containing four computing cells. The CIM macros in the CIM array are interconnected via word line (WL) and bit line (BL). When reading, a non-destructive read voltage is applied between the BL and source line (SL) to generate the output current, while only a single WL is activated at a time. A voltage sense amplifier (VSA) [21] is used for sensing and amplification. The VSA reads the current from an RRAM cell by first converting it into a measurable voltage signal. This is achieved through a charge-discharge mechanism that translates the cell's resistance state into a voltage drop rate on the BL, which is then amplified into a full rail-to-rail digital output without requiring complex multi-stage amplification or external reference sources. After data reading, 12 area-optimized 4T-XOR logic units (Fig. 3(c)) enable multiplication between BF16 input data and 1-bit weights in one clock cycle, achieving a 12× throughput improvement over bit-serial analog CIM methods. Fig. 3(a) shows a computing demonstration of the BF16×1-bit CIM macro. As shown in Fig. 3(c), the implemented XOR circuit features a compact 4-transistor structure that provides efficient logic computation with low power consumption.

Conventional BF16 computations typically use layer-wise alignment, which employs a unified exponent baseline per layer. Although simple, this method can cause significant precision loss and increased hardware overhead when large dynamic ranges within a layer lead to substantial truncation of the mantissa [20]. To overcome these limitations, this work introduces an exponent pre-alignment strategy with fine-grained mantissa handling for CIM architectures. BF16 inputs are pre-aligned in batch-wise using 12-bit expanded mantissas in two's complement format. This allows accumulation without repeated shifting, after which the results are converted back to BF16. As shown in Fig. 3(d), this batch-wise approach markedly reduces shifting errors and improves precision over layer-wise methods.

IV. Evaluation Results

A. Simulation settings

The circuit simulation environment was set up using Cadence Virtuoso, employing the TSMC 28nm process technology. The RRAM devices used in the experiment exhibit a 10% variation with LRS and HRS values of 10.511 kΩ and 321.43 kΩ, respectively. Synthesis of the other digital modules was performed using Design Compiler, while power analysis was conducted using Prime Time. For simplicity, a single PE with the size of 128×128 is used to test the system performance.

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TABLE I. Performance comparison with prior works

Parameter	JSSC'20 [25]	TCAS-I'20 [13]	ESSCIRC'21 [26]	TCAS-II'23 [21]	This Work
Tech. Node (nm)	22	45	130	28	28
RAM Type	RRAM	RRAM	RRAM	RRAM	RRAM
Approach	Analog	Analog	Analog	Digital	Digital
Memory (Mb)	4	0.0625	0.155	0.015625	4
Freq. (MHz)	6.67-200	25	20	125	400
Precision	2/4/8 bit	8-bit	2-bit	1-bit×1-bit	BF16×1-bit
Norm. Eff. (TOPS/W)	391.4	39.04	156.8	1966	442.42

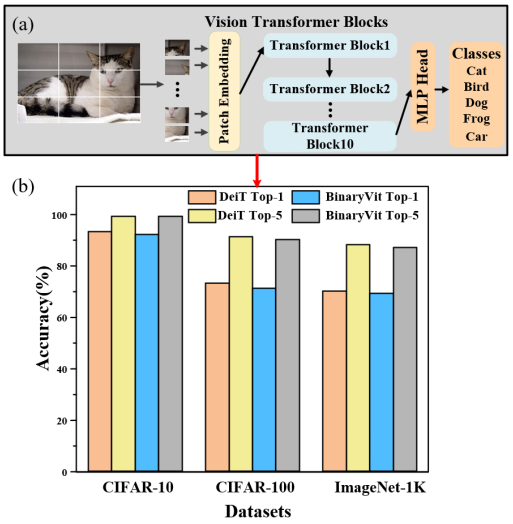


Fig. 5. (a)Image processing pipeline (b)Accuracy of models for different datasets.

B. Accuracy Analysis

Employing a knowledge distillation framework described in Section II, we trained the BF16×1 ViT on CIFAR-10, CIFAR-100, and ImageNet-1K, leveraging a pre-trained DeiT model as the teacher. Fig. 5 summarizes the processing pipeline and the final accuracy results for the quantized ViT and its teacher model (DeiT). The minimal accuracy gap of only 1%-2% between the teacher and student models demonstrates a highly effective knowledge distillation process.

A 128×128 weight matrix was selected from the quantized weights of the ViT, and a randomly generated 128×128 input matrix was used for comparison of software and hardware calculations. After testing, a 128×128 result matrix was obtained, with a mean absolute error of 0.009 and a mean relative error of 0.004. The extremely small errors demonstrate that the system can reliably support real-world network data computation.

C. System Performance Analysis

Based on simulation results of CIM cells and digital modules, the area and power breakdown of the proposed RRAM-based CIM system is shown in Fig. 6. Power is dominated by the accumulator (65%) and peripheral circuits (28.3%), while area is primarily occupied by the RRAM XOR array (53%) and accumulator (26%). Key optimizations include a pre-alignment scheme for floating-point MAC that reduces power through batch processing, and a highly parallel RRAM XOR array that increases computational throughput.

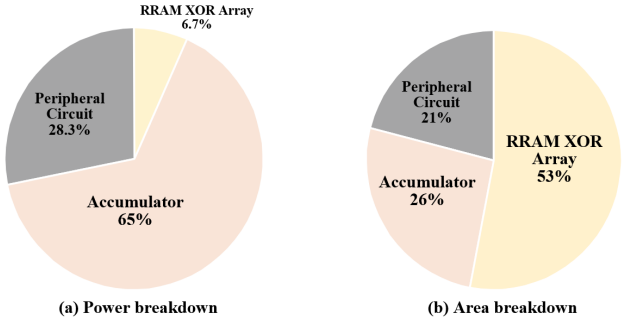


Fig. 6. (a) Power breakdown of proposed design (b) Area breakdown of proposed design

When all normalized to 1-bit, the accelerator achieves a peak energy efficiency of 442.42 TOPS/W on the quantized ViT, outperforming prior works [25], [26], [13] by 1.13×, 2.82×, and 11.33×, respectively. Digital accumulation enables higher operating frequencies without analog limitations. Although the present design consumes more power in the adder tree compared to [21], which uses sparsification for the adder tree, our architecture offers greater generality across various networks, whereas their high efficiency is restricted to particular models.

V. Conclusion

In this paper, we present an RRAM-based digital CIM design for ViT. A novel quantization method is proposed to create BF16×1-bit ViT models, which achieves an optimal balance between memory overhead and computing accuracy. Image recognition tasks using three datasets are implemented on our quantized model with minimal accuracy degradation compared with its full-precision teacher model. Implemented at the 28nm technology node, the proposed design achieves a high power efficiency, measured at 442.42 TOPS/W, while the error is less than 1% compared to the results of the software computation.

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